

ME 316 : Automotive Engine Design

Course Code : ME 316

Course Name : Automotive Engine Design

L-P-T-C: 3-0-0-3

Intended for: UG 3rd, 4th year

Prerequisites: IC140 Graphics for Design

Mutual Exclusion: None

Approval : 57th BoA

Course Contents

- **Understanding General Engine Specification:** Explanation of the engine specification of a typical 2-wheeler and a 4-wheeler engine. Fundamental difference between SI and CI Engines. Correlation of piston movement and air standard cycles for 2 Stroke & and 4 Stroke Engines. Understanding the impact of engine displacement, bore, stroke, and bore-to-stroke ratio. (6 Hours)
- **Head & Block Design Combustion chamber design:** SI, DISI, IDI, CI, etc. Valve port and manifold design. Understanding swirl, tumble, supercharging, and turbocharging in engines. Engine block design for air cooled and water-cooled engine. Understanding design and manufacturing of cylinder liner. Learning various fuel injection systems: PFI, GDI, Carburetor, etc. (10 Hours)
- **Crank Train Design:** Definition of crank train function and terminology. Common crank train configurations. Design of piston, piston rings, connecting rods, crankshafts, balancer shaft, and flywheel, Dynamic balancing and configuring engine firing order. (12 Hours)
- **Valve Train Design:** Overview of valve train. Design and manufacturing of camshaft. Current status of valvetrains VVT, 4-Valve, 3-Valve, 2-Valve, etc. Dynamics of valve train. (6 Hours)
- **Gear Train Design:** Definition of typical gear trains: CVT, DSG, DCT, etc. Function of Clutch. Working of gear shifting mechanism. Design/selection of gears and bearings. Lubrication and oil pump in a single-cylinder engine. (8 Hours)

Text books:

1. Kevin Hoag & Brian Dondlinger, **Vehicular Engine Design**, Springer
2. A Kolchin & V Demidov, **Design of Automotive Engines**, MIR Publishers

References:

1. **Diesel-Engine Management**, Robert Bosch GmbH
2. **Gasoline-Engine Management**, Robert Bosch GmbH
3. JB Heywood, **Internal Combustion Engines**, McGraw Hill.
4. V Ganeshan, **Internal Combustion Engines**, McGraw Hill.
5. William H Crouse, **McGraw Hills Automotive Engine Design**, McGraw Hills